

Milestone - 4

ExoHabitAI – Project Deployment Documentation

1. Objective

The objective of this deployment is to make the ExoHabitAI application publicly accessible through a web URL.

The deployed system integrates:

- Trained Machine Learning Model
- Flask Backend API
- Frontend User Interface
- Production Hosting Environment

2. Deployment Overview

The application is deployed using:

- Hugging Face Spaces (Primary Deployment Platform)
- GitHub (Source Code Repository)
- Hugging Face Model Hub (Model Hosting)

3. Backend Preparation for Deployment

Step 1: Flask App Configuration

The app.py file contains:

- Flask app initialization
- CORS configuration
- API routes (/predict, /rank)
- JSON-based request and response handling
- Dynamic model loading

Step 2: Dynamic Model Loading

To avoid committing large model files, the trained model is dynamically downloaded at runtime using Hugging Face Model Hub which ensures:

- GitHub repo remains lightweight
- Model is always available at runtime
- No manual file management on the server

4. Frontend–Backend Integration

The frontend communicates with the backend using same-origin requests:

```
const API = window.location.origin;
```

This allows:

- Seamless local development
- Seamless deployment
- No hardcoded URLs
- No CORS issues

5. Updating requirements.txt

The following dependencies are included:

```
flask
flask-cors
numpy
pandas
scikit-learn
joblib
huggingface_hub
gunicorn
```

All required libraries for model loading, prediction, and serving are covered.

6. Deployment on Hugging Face Spaces

- The application was deployed on Hugging Face Spaces using a Flask backend.
- The repository was linked, and app.py was used as the entry point.
- The environment was automatically built and served on port 7860.
- The trained model was loaded dynamically from the Hugging Face Hub, avoiding large file storage in the repository.
- The deployed application was made available via a public web URL.

7. Environment Variables

No environment variables are required since:

- Model is fetched dynamically
- No API keys are hardcoded
- All paths are relative and platform-safe

8. Testing the Deployed Application

After deployment, the public URL is generated:

<https://janhvi01-exohabitai.hf.space>

The following tests were performed:

- Homepage loads successfully
- Single planet prediction works
- JSON-based prediction works
- Multi-planet ranking works
- Charts and UI render correctly
- No backend crashes observed

9. Errors Encountered and their fixes

- Issue: Prediction failed after deployment
Cause: Frontend was calling localhost
Fix: window.location.origin was used for API calls
- Issue: Model file not found
Cause: Model excluded via .gitignore
Fix: Model was uploaded to Hugging Face Model Hub and loaded dynamically
- Issue: CORS errors
Fix: Same-origin deployment + Flask-CORS

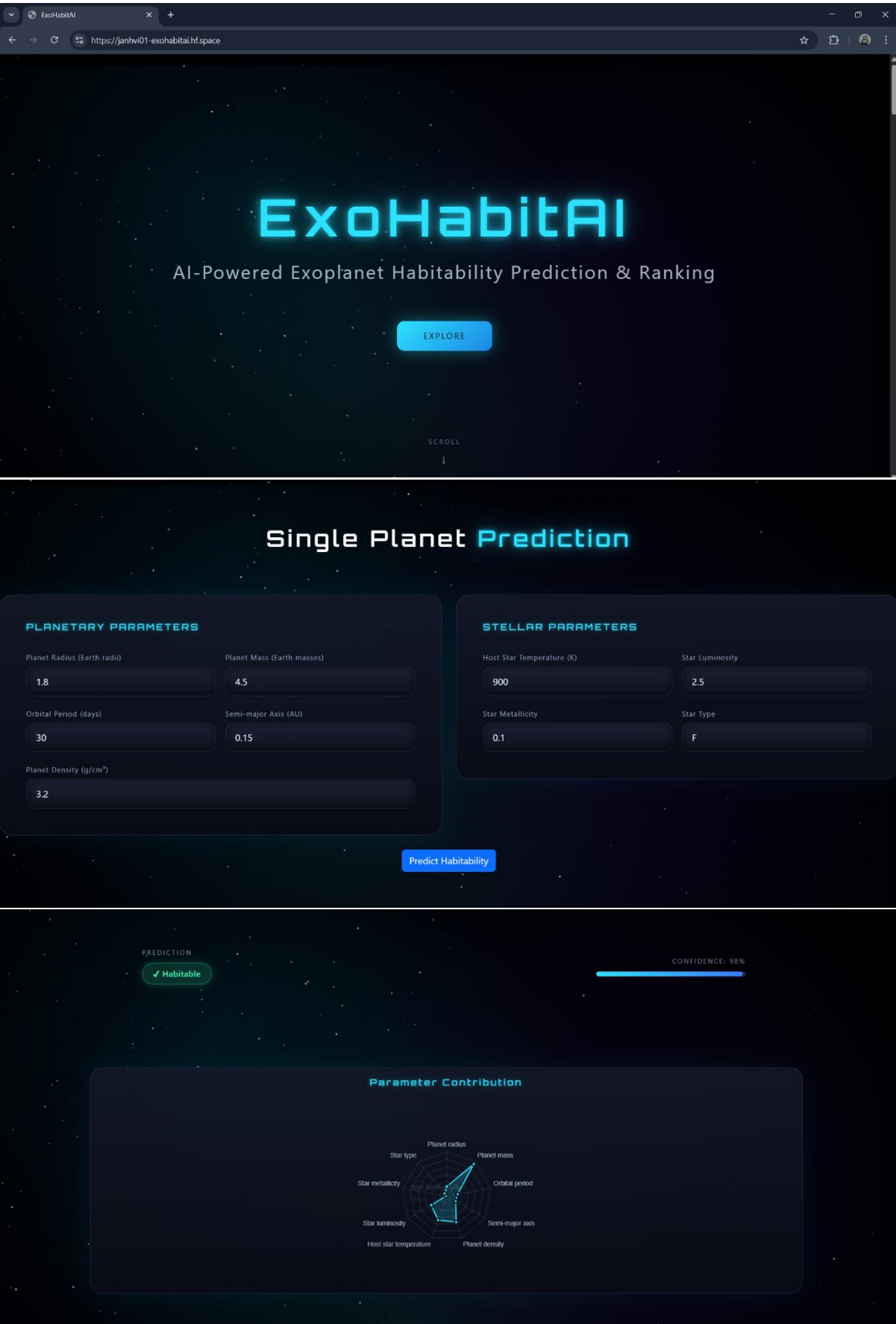
11. Deployment URL

Live Application: <https://janhvi01-exohabitai.hf.space>

Conclusion

The final outcome is a fully functional AI-powered exoplanet habitability prediction system accessible via a browser.

12. Working Web App Snapshots



JSON Input

```
{  
  "Planet radius": 1.1,  
  "Planet mass": 1.8,  
  "Orbital period": 380,  
  "Semi-major axis": 1.05,  
  "Planet density": 5.2,  
  "Host star temperature": 300  
}
```

Predict from JSON

PREDICTION

Not Habitable

CONFIDENCE: 39%

Parameter Contribution



Multi-Planet Ranking

```
{  
  "Semi-major axis": 0.15,  
  "Planet density": 3.2,  
  "Host star temperature": 900,  
  "Star luminosity": 2.5,  
  "Star metallicity": 0.1,  
  "Star type": "F"  
}
```

```
{  
  "Planet density": 5.5,  
  "Host star temperature": 288,  
  "Star luminosity": 1.0,  
  "Star metallicity": 0.02,  
  "Star type": "G"  
}
```

```
{
  "Semi-major axis": 2.0,
  "Planet density": 1.2,
  "Host star temperature": 300,
  "Star luminosity": 1.2,
  "Star metallicity": 0.0,
  "Star type": "G"
}
```

```
{
  "Semi-major axis": 1.05,
  "Planet density": 5.2,
  "Host star temperature": 290,
  "Star luminosity": 1.0,
  "Star metallicity": 0.02,
  "Star type": "G"
}
```

```
{
```

```
{
  "Planet radius": 2.8,
  "Planet mass": 12,
  "Orbital period": 80,
  "Semi-major axis": 0.3,
  "Planet density": 2.0,
  "Host star temperature": 850,
}
```

Add Exoplanet

Rank Exoplanets

#1 Planet 1 98%

#2 Planet 2 54%

#3 Planet 3 39%

#4 Planet 4 0%

#4 Planet 4 0%

#5 Planet 5 0%

Habitability Score Comparison

